KIMODO-POLICY: FROM TEXT-TO-MOTION GENERATORS TO HUMANOID VISION-LANGUAGE-ACTION POLICIES

Anonymous authors
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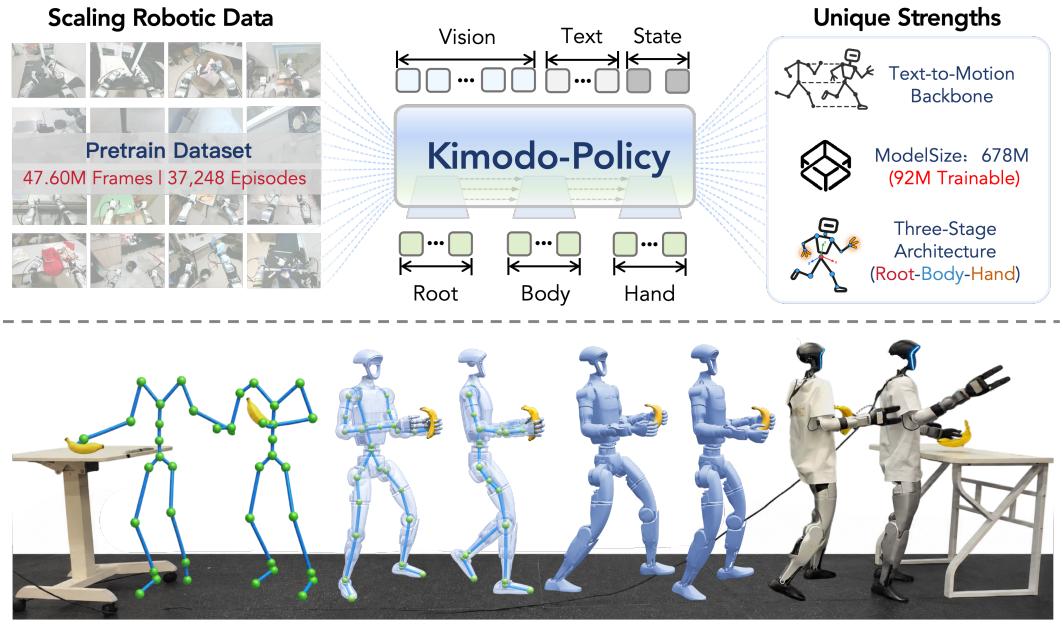


Figure 1: **Overview of Kimodo-Policy.** Top: diverse VLA pretraining adapts T2M priors through three-stage root–body–hand generation for visually grounded control. Bottom: skeletal history conditions action-chunk prediction; real-world examples illustrate coordinated whole-body execution.

ABSTRACT

Humanoid vision–language–action (VLA) models translate multimodal task understanding into coordinated whole-body behavior. Existing approaches equip pretrained vision–language or video generation backbones with action experts, but their pretraining does not directly establish whole-body motion priors. Unified or upper–lower-body action designs also lack dedicated root, body, and hand modeling despite their distinct requirements. These limitations increase the burden of learning precise, coordinated actions from limited task-specific humanoid demonstrations. We introduce Kimodo-Policy, which initializes visually grounded control with pretrained text-to-motion (T2M) priors over inter-joint coordination and temporal structure. Its three-stage root–body–hand architecture combines dedicated generators with explicit dependencies: root trajectories condition body generation, while future body features and visual observations guide hand control. We pretrain the policy on diverse humanoid demonstrations, conditioning actions on visual observations, language instructions, and motion history. With 678M total and only 92M trainable parameters, Kimodo-Policy achieves strong performance on HumanoidArena, SIMPLE, and real-world loco-manipulation tasks. Further experiments show performance gains from larger VLA pretraining datasets, greater model capacity, and stronger pretrained T2M priors.

1 INTRODUCTION

Vision–language–action (VLA) models integrate visual perception, language understanding, and action generation across manipulation tasks and robot embodiments (Brohan et al., 2023; Kim et al., 2024; Black et al., 2025b; Wang et al., 2026b; 2025). Learning instruction-conditioned behaviors from demonstrations provides a unified mapping from multimodal inputs to robot actions. Extending VLA learning to humanoids could broaden task coverage in human environments, where locomotion, posture, and manipulation must be coordinated (Bjorck et al., 2025; Wei et al., 2026a; Hu et al., 2026). For example, transferring objects between tables requires grasping, turning, walking, and placement while maintaining balance (Fig. 1, bottom). Such tasks require grounding task intent and visual feedback in coherent whole-body actions, motivating the extension of robot policy architectures to humanoid loco-manipulation.

Humanoid policy learning commonly extends the backbone–action expert paradigm from manipulation to whole-body control. Pretrained vision–language models encode instruction and observation semantics (Fig. 2(a)) (Bjorck et al., 2025; Wei et al., 2026a; Hu et al., 2026), while video generators supply spatiotemporal priors for scene evolution (Fig. 2(b)) (Zheng et al., 2026; Li et al., 2026). These representations condition action generation across root, body, and hand motion. Unified predictors generate whole-body sequences (Fig. 2(d)) (Hu et al., 2026; Zheng et al., 2026); hybrid designs combine upper-body joint targets with lower-body locomotion commands executed by dedicated controllers (Fig. 2(e)) (Zhong et al., 2025; Jiang et al., 2025). The former captures cross-component dependencies; the latter couples upper-body control with learned locomotion and stabilization. Both transfer pretrained semantic and visual dynamics knowledge to humanoid loco-manipulation.

Despite these advances, two challenges remain. ❶ **Insufficient Whole-Body Motion Priors.** Compared with fixed-base manipulation, humanoid control requires higher-dimensional inter-joint and temporal coordination. Backbone pretraining (Bjorck et al., 2025; Wei et al., 2026a; Zheng et al., 2026; Li et al., 2026) does not directly equip action generators with priors aligned with target humanoid kinematics, leaving them to learn these priors, often from limited task-specific demonstrations. ❷ **Insufficient Differentiation of Motion Components.** Root, body, and hand control differ in scale and precision. Unified prediction (Hu et al., 2026; Zheng et al., 2026) may create competing objectives; hybrid execution (Zhong et al., 2025; Jiang et al., 2025) does not itself provide dedicated root, body, and hand generators. Both must accommodate component-specific requirements while maintaining whole-body coordination.

Recent text-to-motion (T2M) methods learn structured motion distributions conditioned on text or spatial constraints (Petrovich et al., 2022; Tevet et al., 2023; Chen et al., 2023; Zhang et al., 2023a; Guo et al., 2024; Xie et al., 2024). Building on these advances, **Kimodo-Policy** combines a pre-

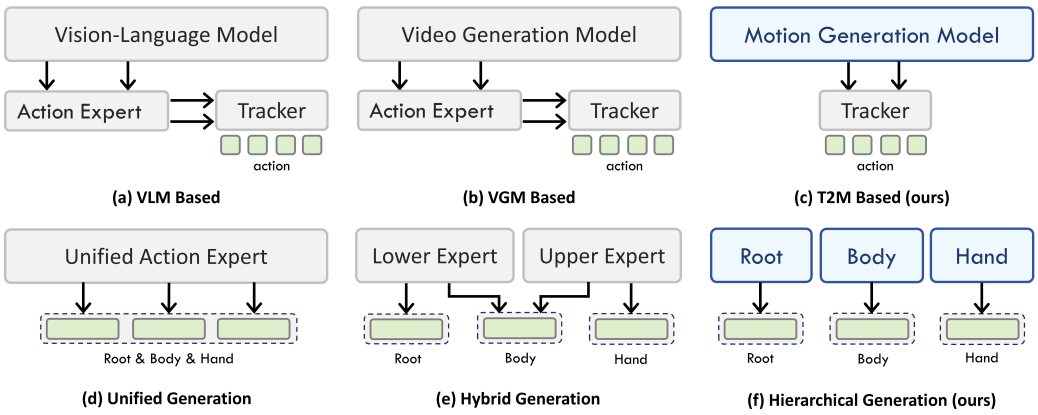


Figure 2: **Comparison of Humanoid Policy Architectures.** Panels (a)–(c) contrast VLM-based, VGM-based, and our T2M-based policy backbones. Panels (d)–(f) illustrate unified whole-body generation, hybrid generation of upper-body joint targets and lower-body locomotion commands, and our hierarchical root–body–hand generation, respectively.

trained T2M backbone with three-stage root–body–hand generation for humanoid VLA control. ❶ **Strong Whole-Body Motion Priors.** We transfer Kimodo's pretrained inter-joint and temporal priors to high-dimensional, visually grounded control (Fig. 2(c)) (Rempe et al., 2026), easing motion learning from limited task-specific humanoid demonstrations. ❷ **Strong Component-Specific Motion Modeling.** Dedicated generators (Fig. 2(f)) predict root translation and heading, root-conditioned body motion, and hand controls conditioned on future body features and visual observations. Their dependencies support component specialization and whole-body coordination. We pretrain on diverse humanoid demonstrations to generate actions conditioned on visual observations, language instructions, and motion history. With 678M total and only 92M trainable parameters, Kimodo-Policy shows consistent gains from more VLA pretraining data, larger models, and stronger T2M priors. Our main contributions are:

- We introduce **Kimodo-Policy**, a humanoid VLA framework transferring pretrained T2M priors to whole-body control. Pretraining on diverse humanoid demonstrations conditions closed-loop actions on visual observations, language instructions, and motion history.
- We propose dedicated root, body, and hand generators in a three-stage architecture to meet distinct control requirements. Explicit cross-stage conditioning preserves motion dependencies, supporting specialized motion generation and coordinated whole-body control.
- We demonstrate strong performance on HumanoidArena, SIMPLE, and real-world humanoid tasks using 678M parameters (only 92M trainable), with consistent gains from larger VLA pretraining datasets, greater model capacity, and stronger T2M priors.

2 Related Work

2.1 Humanoid Vision-Language-Action Models

Humanoid VLA models typically use pretrained vision–language models (VLMs) or video generation models (VGMs). VLM-based methods transfer visual-semantic knowledge to actions (Bjorck et al., 2025; Wei et al., 2026a; Zhong et al., 2025; Hu et al., 2026): GR00T N1 (Bjorck et al., 2025) couples a VLM with a diffusion action model, while Ψ_0 (Wei et al., 2026a) transfers representations from human egocentric demonstrations to a humanoid action expert. VGM-based methods exploit visual dynamics priors (Zheng et al., 2026; Li et al., 2026): MotionWAM (Zheng et al., 2026) generates whole-body motion from intermediate video features; WholeBodyWAM (Li et al., 2026) grounds world–action priors in whole-body controllers. Unified generators (Hu et al., 2026; Zheng et al., 2026) share prediction across motion components; hybrids (Zhong et al., 2025; Jiang et al., 2025) combine upper-body targets with locomotion commands. HAF (Gu et al., 2026) stages generation with a shared action expert, expanding the active action set and refining earlier predictions. Kimodo-Policy instead transfers a T2M backbone with pretrained inter-joint and temporal priors, reducing the need to learn motion regularities from humanoid demonstrations. Dedicated root, body, and hand generators meet distinct control requirements, while cross-stage conditioning supports coordinated, visually grounded whole-body behavior.

2.2 Text-to-Motion (T2M)

T2M includes TEMOS's variational language–motion modeling (Petrovich et al., 2022), pose-space MDM (Tevet et al., 2023) and latent-space MLD (Chen et al., 2023) diffusion, and autoregressive T2M-GPT (Zhang et al., 2023a) and masked MoMask (Guo et al., 2024) quantized token generation. OmniControl (Xie et al., 2024) adds spatial constraints; Kimodo (Rempe et al., 2026) trains controllable root–body diffusion hierarchies at scale for human and humanoid skeletons. Humanoid-VLA (Ding et al., 2025) and ZeroWBC (Yang et al., 2026) learn autoregressive motion priors in pretrained language models and VLMs before visual integration and task adaptation, respectively. MotionWAM (Zheng et al., 2026) adapts a pretrained video world model via staged video–motion training; WholeBodyWAM (Zhang et al., 2026) integrates a pretrained motion expert with video and action experts. Kimodo-Policy directly reuses a diffusion T2M backbone pretrained at scale in robot kinematic space. Dedicated root–body–hand generators ground inter-joint and temporal priors in vision and motion history for closed-loop loco-manipulation, reducing relearning from robot demonstrations while meeting distinct control requirements.